

OLGA SOROKOLETOVA

Mechanistic Interpretability for AI Safety

✉ sorokoletova@diag.uniroma1.it
📍 Rome, Italy

🌐 olgasorokoletova

👤 olga-sorokoletova

🎓 Google Scholar

🆔 0009-0005-4356-2649

CURRENT ROLE

PhD Researcher — Mechanistic Interpretability for LLM Adversarial Defence

University of Rome La Sapienza, Dept. of Computer, Control and Management Engineering

📅 Nov 2025 – Present

📍 Rome, Italy

- National PhD in Artificial Intelligence; advisor Prof. Daniele Nardi. Scholarship funded by Accredia, with a research remit on technical standardisation for accuracy, robustness, and cybersecurity of high-risk AI systems under the EU AI Act, and on international standardisation for generative AI. Member of ICARO Lab.
- **Why literary jailbreaks evade safety training**  **arXiv Preprint:** [5]
First and corresponding author. Follow-up to an ICARO Lab finding that poetic reformulation acts as a universal jailbreak [6], on which I was a co-author; this paper explains the mechanism. On Qwen3-14B (46.9% vs. 10.4% attack success, poetry vs. prose): attention patterns decode literary format near-perfectly (0.985 acc., 0.997 AUC) yet predict jailbreak success barely above chance (≈ 0.66). Format recognition and harm detection are therefore decoupled, locating safety-relevant computation downstream of attention. Device-level ablation shows no single poetic device is necessary or sufficient: jailbreaks arise from accumulated stylistic irregularity plus metaphorical substitution of post-training lexical triggers. Contributed a reusable 72-d interpretable attention representation (generation phase $\times$ layer cluster $\times$ functional token group). Extracted ~ 600 GB of attention tensors on a single 24GB GPU across 2,397 prompts; three-judge open-weight ensemble with human validation; seeds fixed throughout. Under review. *Ongoing*: testing whether a “poetic processing mode” forms an identifiable direction in activation space, and whether steering along it attenuates the attack.
- **Recoverability: can an LLM restore its own alignment after a safety failure?**  **Workshop Publication:** [2]
Proposed a shift in framing: rather than building models that never misalign, treat *self-recovery* as a measurable dimension of safety in its own right. Introduced a trajectory-based methodology for modelling alignment dynamics across multi-turn adversarial dialogue, formalised recovery trends (absolute vs. temporary; recovery paths), and defined two metrics, Misalignment Length and Recovery Duration. Analysis over 597 adversarial dialogues (7,635 messages, 2,454 turns, manually annotated): recovery is rare (14.6% of conversations; 3.2% recover more than once) and risk-dependent, with well-bounded hazard categories recovering better than context-dependent ones. Also showed that the choice of moderation model materially changes the measurement: a larger Llama Guard agreed *less* with human ground truth than a smaller one (69.5% vs. 77.1%). Project page. *Ongoing*: testing whether targeted mechanistic interventions can shorten misalignment and extend recovery.
- Also: LLM-as-a-judge for gender stereotype detection in natural language texts. 1st and 2nd place, GSI:detect shared task, EVALITA 2026 [3]. Structured prompting and fine-tuning for joint detection and classification.

PREVIOUS RESEARCH EXPERIENCE

Research Fellow — LLM Adversarial Robustness

University of Rome La Sapienza, Dept. of Computer, Control and Management Engineering

📅 Nov 2024 – Oct 2025

📍 Rome, Italy

- **Designed and ran a red-teaming challenge on generative AI safety** as a graded component of the MSc course *Seminars in Artificial Intelligence and Robotics*, with 48 enrolled students as red-teams. Collected 1,364 adversarial multi-turn dialogues against Minerva-7B-instruct-v1.0 in a black-box setting, with attacks unrestricted in form: single- and multi-turn, human-crafted and optimisation-based, multilingual, conducted primarily in Italian. Every interaction was manually annotated for attack success and technique used. The resulting corpus underpins two subsequent papers.
- **A mechanism-oriented taxonomy of jailbreak techniques**  **Conference Publication** [1]
Second author. Built a three-level taxonomy organising 50 jailbreaking techniques into seven mechanism-oriented families defined by how a prompt bypasses safeguards: impersonation, persuasion, privilege escalation, cognitive overload, obfuscation, goal conflict, and data poisoning. The contribution is a consolidation rather than a novelty, harmonising techniques that prior work studied in isolation, with explicit cross-references to existing classifications, and then refining them against attacks observed during the challenge. Analysed prevalence and success rates across families (185 successful attacks of 1,364); a preliminary experiment confirmed that taxonomy-guided prompting improves GPT-5's jailbreak detection. Released the first Italian jailbreak dataset, annotated by technique rather than risk category and covering multi-turn attacks where adversarial intent emerges gradually.

WORK EXPERIENCE

Machine Learning Research Engineer — NLP

CY4GATE, Dept. of Data and AI

📅 Oct 2023 – Oct 2024

📍 Rome, Italy

- **0-CTI: Zero-shot Cyber Threat Intelligence extraction**

📄 **Workshop Publication:** [4]

Designed a modular, STIX-compliant framework that turns unstructured cyber threat reports into a knowledge graph over 9 entity and 21 relation types, running fully zero-shot or supervised where labels exist. Benchmarked head-to-head against STIXnet, the prior state of the art in CTI entity extraction: overall F1 0.98 vs. 0.77, ahead on all seven entity types compared; 0.85 weighted test F1 on a more diverse OpenCTI-sourced corpus (9 classes, 34k annotated entities). Contributed a zero-shot relation-extraction algorithm recasting the task as entailment scoring, generating candidate triples from the STIX schema and scoring them against the source text with a pretrained cross-encoder. Zero-shot components assessed qualitatively by LLM-as-a-judge over 120 reports: mean 0.91 (SD 0.06) for entities, 0.83 (SD 0.15) for relations. Integrated into the company's Decision Intelligence platform and applied to attack-model enrichment for [Alnception](#), an international cyber-defence project.

Risk Analyst

OTP Bank, Dept. of Risk Management

📅 Sep 2019 – Sep 2020

📍 Moscow, Russia

- Developed a deep-learning credit-scoring model that outperformed the bank's existing method, and automated NPV calculation across the loan portfolio.
 - Statistical segmentation of COVID-affected client cohorts to sustain lending under elevated default risk.
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EDUCATION

Certificate of Advanced Specialization in Generative AI

Nebius Academy, School of AI and Data Technologies

📅 2024

📍 Online

18-week intensive completed alongside full-time work, taken to build LLM depth beyond my NLP engineering background: language modelling, attention and transformer architecture, RLHF and DPO, distributed training, efficient inference, fine-tuning, multimodality, evaluation, agent design, AI safety, and image generation (VAEs, GANs, diffusion).

Course project: built a domain-specific knowledge base and multi-turn RAG pipeline, benchmarking reranker integration, embedding-model retrieval differences, and output variance across LLM APIs, with an LLM-as-judge evaluation harness.

M.Sc. in Artificial Intelligence and Robotics

University of Rome La Sapienza, Dept. of Computer, Control and Management Engineering

📅 2020 – 2023

📍 Rome, Italy

110/110 (maximum). Thesis: *Enhancing Robot Reliability for Health-Care Facilities by means of Human-Aware Navigation Planning*, in collaboration with LAAS-CNRS (Toulouse).

📄 **arXiv Preprint:** [7]

B.Sc. in Applied Mathematics and Computer Science

M.V. Lomonosov Moscow State University, Dept. of Computational Mathematics and Cybernetics

📅 2015 – 2019

📍 Moscow, Russia

4.7/5.0. Thesis: *Random Sampling Method for Cryptocurrency Market Time Series Forecasting*.

📄 **Journal Publication:** [8]

PUBLICATIONS

Author's name in **bold**.

Conference Publications

[1] F. Giarrusso, **O. E. Sorokoletova**, V. Suriani, and D. Nardi, "Guarding the Guardrails: A Taxonomy-Driven Approach to Jailbreak Detection," in *Proceedings of IASEAI Conference*, vol. 2, 2026, pp. 191–203. [Online]. Available: <https://ojs.aaai.org/index.php/IASEAI/article/view/43024>.

Workshop Publications

[2] **O. E. Sorokoletova**, F. Giarrusso, V. Suriani, and D. Nardi, "Learning from Mistakes: Can LLM Self-Recover after Misalignment?" In *Machine Ethics: from formal methods to emergent machine ethics Workshop (MEW2026) co-located with the 40th Annual AAAI Conference on Artificial Intelligence (AAAI 2026)*, vol. 4189, 2026, pp. 74–83. [Online]. Available: <https://ceur-ws.org/Vol-4189/paper5.pdf>.

[3] **O. E. Sorokoletova**, E. Musumeci, and D. Nardi, "DIAG-Sapienza at GSI: Detect: Joint Detection and Classification of Gender Stereotypes with Structured Prompting and Fine-Tuning,," in *Proceedings of the 9th Evaluation Campaign of Natural Language Processing and Speech Tools for Italian. Final Workshop (EVALITA 2026)*, vol. 4195, 2026. [Online]. Available: <https://ceur-ws.org/Vol-4195/43.pdf>.

[4] **O. E. Sorokoletova**, E. Antonioni, and G. Colò, "Towards a Scalable AI-Driven Framework for Data-Independent Cyber Threat Intelligence Information Extraction," in *Proceedings of the First International Workshop on Generative AI for Textual Document Analysis*, 2024. DOI: 10.1109/FLLM63129.2024.10852465.

arXiv Preprints

[5] **O. E. Sorokoletova**, F. Giarrusso, G. D. Luca, P. Bisconti, M. Prandi, F. Pierucci, M. Galisai, V. Suriani, et al., "Metaphor Is Not All Attention Needs," 2026. [Online]. Available: <https://arxiv.org/abs/2605.12128>.

[6] P. Bisconti, M. Prandi, F. Pierucci, F. Giarrusso, M. B. Syrnikov, M. Galisai, V. Suriani, **O. E. Sorokoletova**, et al., "Adversarial Poetry as a Universal Single-Turn Jailbreak Mechanism in Large Language Models," 2025. [Online]. Available: <https://arxiv.org/abs/2511.15304>.

[7] **O. E. Sorokoletova** and L. Iocchi, "Enhancing Robot Reliability for Health-Care Facilities by means of Human-Aware Navigation Planning," 2023. [Online]. Available: <http://arxiv.org/abs/2409.17131>.

Journal Publications

[8] **O. E. Sorokoletova** and T. V. Zakharova, "Random Sampling Method for Cryptocurrency Market Time Series Forecasting," *Systems and Means of Informatics*, vol. 29, pp. 65–72, 4 2019.

TECHNICAL SKILLS

Interpretability & Analysis

Attention Pattern Analysis

Probing Classifiers

Ablation Studies

Feature Design for Interpretability

LLM Engineering

PyTorch

Structured Prompting

LLM-as-a-Judge

Fine-Tuning

RAG

HuggingFace Transformers

Open-Weight Model Deployment

Data & Experimentation

Weights & Biases

Label Studio

NumPy

Pandas

SciPy

scikit-learn

Matplotlib

Seaborn

Multi-Annotator Pipelines

Infrastructure

Remote GPU Workflows

Linux

SSH/rsync/rcclone

Docker

Git

Azure

FastAPI

Languages & Other Tooling

Python

C/C++

SQL

MATLAB

spaCy

RESEARCH AREAS

Mechanistic Interpretability

AI Safety & Alignment

Adversarial Robustness

Jailbreaks & Red-Teaming

Multilingual Safety Evaluation

Large Language Models

Natural Language Processing

AI for Cybersecurity

LANGUAGES

Russian

English

Italian